

VisaPath: Project Brief

The Problem

There are over 1 million international students in the US, and every one of them has to navigate a complicated web of visa deadlines, work authorization rules, and immigration paperwork. The consequences of missing a single deadline can be devastating: loss of legal status, inability to work, or even deportation.

Right now, students piece together information from USCIS.gov, Reddit threads, university DSO offices, and immigration lawyers who charge \$300+ per hour. There is no single tool that takes a student's specific situation and gives them a clear picture of what they need to do and when.

The Solution

VisaPath is an AI powered web app that generates a personalized immigration timeline for each student. After a short onboarding (visa type, program details, career goals), the app builds a complete roadmap covering:

Timeline with every critical deadline and milestone, color coded by urgency and grouped by month. Each event expands to show detailed descriptions and action items.

Risk Alerts that flag things like CPT overuse (which kills OPT eligibility), India/China green card backlogs with 10-30+ year waits, and H-1B lottery odds based on wage level.

AI Chat grounded in actual USCIS documentation through RAG (Retrieval Augmented Generation). We embedded 8 immigration knowledge base documents into a vector database so answers cite real sources instead of hallucinating.

Document Tracker with step by step checklists for OPT, STEM OPT, H-1B, and Green Card filings.

Tax Guide personalized to your visa type, country, treaty benefits, and residency status.

What-If Simulator to explore how changing one variable (graduation date, STEM status, country) shifts your entire timeline.

Architecture

```
React 19 + TypeScript + Tailwind CSS v4
  |   REST API (JSON)
  v
Python 3.13 + FastAPI
  |
  |-- 56 hardcoded USCIS rules + filing fees --> Gemini 2.5 Flash --> Structured JSON tim
```

```
|-- 8 immigration docs --> ChromaDB (21 chunks) --> RAG chat
|-- Risk analyzer --> Country backlogs + lottery stats
|-- JWT auth --> SQLite (dev) / PostgreSQL (prod)
```

Key technical decisions: - 56 hardcoded USCIS rules ensure the AI prompt has accurate, up to date information instead of relying on the model's training data - RAG grounds chat answers in official documentation so the AI cites real sources - Client side date recalculation means cached timelines never show stale urgency levels - The frontend navigates to the timeline immediately and shows an animated loading screen while AI generates in the background

Impact

- Prevents missed deadlines that could end a student's legal status
- Makes immigration planning accessible to students who can't afford lawyers
- Reduces anxiety by giving a clear visual roadmap of what's ahead
- Saves hours of research that students currently spend across 10+ different sources

Team

Name	Role
Dileep Kumar Sharma	Full stack development, AI integration, deployment

AI Disclosure

In the product: Google Gemini 2.5 Flash (timeline, chat, tax guide) and gemini-embedding-001 (RAG vectors)

During development: Claude Code (Anthropic) was used as a coding assistant

All immigration guidance includes disclaimers that it's general information, not legal advice.

Tech Stack

React 19 / TypeScript / Vite 7 / Tailwind CSS v4 / FastAPI / Python 3.13 / Google Gemini 2.5 Flash / LangChain / ChromaDB / SQLite + PostgreSQL / Azure App Service / GitHub Actions

Links

- Live: visapath-app.azurewebsites.net

- Repo: github.com/dileep-kumar-sharma/visapath
- Video: *[coming soon]*